Theoretical architecture of SMC-BVS v3: conjectured theorems, assumptions and proof routes

Theory-architecture run

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Abstract

Status. Every statement in this document is a conjecture about the exact method of METHODOLOGY_v3.md (version 3.0, 2026-10-19). Nothing is proven. The document presents the 36 conjectured results of the run as a paper would present them: the setting and notation, the assumptions with their discussion, each result with its purpose, and the connections between them. The full statements, with every field needed by a proof run, are in the files `theorems/T01...T36`; this document summarises and explains. Numerical results from the judge's assistants are throwaway evidence and are labelled *provisional* throughout.

What the theory says, in one paragraph. If the target-transfer conjectures hold, the posterior fixed by the method (standardised data, the $g = n$ slab, $\mathrm{IG}(0.01, 0.01)$, the capped $\mathrm{BB}(1, p)$ prior) concentrates on the true support under sparse eigenvalues and a fixed beta-min constant, and contracts for prediction and variance without beta-min. Given exact arithmetic, ideal random streams and a frozen calibration, the final run's evidence and evidence-weighted Rao–Blackwell identities hold exactly; they give no finite-budget accuracy by themselves. The reported HPM is the truth whenever the truth is the strict target mode *and some scoring route finds it*: initial draws, Gibbs evaluations, swap proposals, block tables, local scans, or a terminal repair. PIPs need a terminal occupancy, entrance and stability argument; the evidence needs either a single dominating model along the bridge (a first-moment argument) or particle-system stability through a crossing, which marginal occupancy does not supply. Negative results are conditional comparison theorems: when calibration avoids a hidden set and the run stays coupled to its complement, the evidence is biased by the restricted-mass factor while every evidence-measurable diagnostic looks normal; no production family realising this has been constructed. Finally, each reported certificate has a precise and limited meaning.

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1 Introduction and reading guide

The method under study is SMC-BVS v3, a sequential Monte Carlo sampler for Bayesian variable selection with a composite block-table start, a bridge q_ϕ between the start and the posterior, a pilot that calibrates a temperature grid, a final run of four islands, and an all-scored HPM decision. The run that produced this document followed the theoretical-architecture protocol: a council of four experts conjectured candidate theorems; a review group of four assessed them and sent six fatal-as-stated objections back; the council answered; a judge reviewed the whole proposal, ordered twelve single-seed numerical checks, and amended it; the lead ruled on every amendment (`RULINGS.md`). The result is organised like a theory paper: Part A on the target posterior, Part B on fixed-data Monte Carlo validity, Part C on combined guarantees for the actual output, Part D on limits and negative results.

Three distinctions run through everything and the reader should keep them apart:

- a property of the exact mathematical algorithm (an identity), versus a conditional theorem whose computational hypotheses are unverified, versus an explicitly idealised component (fixed calibration with $N \to \infty$), versus an open question;
- *scoring* a model (which the HPM needs), *visiting* it, and *occupying* it at a recording position (which PIPs and the evidence need);
- discovery of the truth by calibration, and accuracy of the evidence or the PIPs: the first never implies the second.

The decisions fixed by the author of the method are not changed by anything here: the target, the cap $K = \min(p, \lfloor n/2 \rfloor)$, the block rule, $B = \min(\lfloor \sqrt{n} \rfloor, \lfloor \sqrt{p} \rfloor, 16)$, the neighbour threshold $\rho_n = \sqrt{2 \log p / n}$, the HPM decision with exact ties, the single prior arm, and the budgets. Where a result would benefit from a change, the change appears only as a recommendation (Section 9).

2 Setting and notation

Data and target. Raw data (X^r, y^r) with $n \geq 3$; $C = I - \mathbf{1}\mathbf{1}^\top/n$; $X = CX^r D_x^{-1}$ and $y = Cy^r/s_y$ with $s_y^2 = \|Cy^r\|^2/(n-1)$, so every column and y have squared norm $n-1$. Contrast coordinates use $U \in \mathbb{R}^{n \times (n-1)}$ with $U^\top U = I$, $UU^\top = C$. The state space is $\Omega = \{S \subseteq [p] : |S| \leq K\}$. For $S \in \Omega$,

$$G_S = X_S^\top X_S = (n-1)R_S, \qquad z_S = X_S^\top y, \qquad H_S = (n+1)G_S + I,$$

$$D_S = \frac{\det H_S}{\det(G_S + I)}, \qquad Q_S = y^\top y - n z_S^\top H_S^{-1} z_S,$$

$\nu = 0.01 + n/2$, $m(S) \propto D_S^{-1/2}(0.02 + Q_S)^{-\nu}$, $w(k) = \frac{1}{2}\binom{2p-1}{k}^{-1}$, $\varpi(S) = w(|S|)/W_K$, $F(S) = \log[\varpi(S)m(S)]$, $Z = \sum_\Omega \varpi(S)m(S)$, $\pi = e^F/Z$, $\mathrm{BF}(S) = m(S)/m(\varnothing)$. The global HPM $S^\dagger$ maximises F over Ω (ties: smallest size, then lexicographic); the reported HPM $\widehat{S}$ applies the same rule over the set $\mathcal{S}$ of all models ever *scored*. $\mathrm{PIP}_j = \pi(j \in S)$ and $\mathrm{MPM} = \{j : \mathrm{PIP}_j > 1/2\}$.

Statistical model. $y^r = \alpha_* \mathbf{1} + X^r \beta_* + \varepsilon$, $\mu = CX^r\beta_* = X\theta_*$, $S_0 = \operatorname{supp}\beta_*$, $s_0 = |S_0|$. Contrast-scale parameters $\theta = s_y\beta$, $v = s_y^2\sigma^2$. $r_n = \sqrt{\log n/n}$.

Sampler. Deterministic blocks $C_1, \dots, C_J$ of size at most B (complete linkage, top-down cut, next-fit packing); tables $r_b(A) = \mathrm{BF}(A)$ for $A \subseteq C_b$; composite start $\gamma_0(S) = \varpi(S) \prod_b r_b(S \cap C_b)$, $\pi_0 = \gamma_0/\bar{Z}_0$ (exact convolution, exact iid sampler); start mode $\widetilde{S}_0 = \arg\max \pi_0$; discrepancy $d(S) = \log \mathrm{BF}(S) - \sum_b \log r_b(S \cap C_b)$; bridge $q_\phi \propto \gamma_0 e^{\phi d}$, so $q_0 = \pi_0$, $q_1 = \pi$. A round K_ϕ executes $m = p + B + 2$ labelled steps in uniformly random order: p Gibbs flips (with pre-draw inclusion probabilities $o_j(S, \phi)$), one uniform swap, B neighbour swaps, one independence proposal from π_0. The calibration $\mathcal{C}$ (grid $0 = \phi_0 < \cdots < \phi_T = 1$, candidate set V, tier, N_e, L) is fixed by streams 0–9; the final run uses fresh streams: four islands of N_e iid π_0 draws, multinomial resampling after every reweighting, one round per stage, L burn-in and four measured rounds at $\phi = 1$, then a local search of at most ℓ moves whose scans score every feasible flip and swap of the centre. Outputs: island evidences $\widehat{Z}_r$, $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r$, the bound $0.05\widehat{Z}$, Rao–Blackwell PIPs $\widehat{\mathrm{PIP}}_{rj}$ (pre-draw Gibbs conditionals averaged over the measured rounds), primary (equal-weight) and secondary ($\widehat{Z}_r$-weighted) pooled PIPs, $\widehat{S}$, statuses and flags. Budgets: $N_0 = \max(2500, n, p)$, $H_0 = \lceil 512 N_0/2500 \rceil$, $\ell = \lceil \log_2(n + p + 1) \rceil$, $T_0 = \max(1000, \ell^2)$, tier $e \in \{0, 1\}$ multiplies by 4^e, $L = \max(4, \lceil \log_2(n+1) \rceil)$; $Q_f = 4N_e(T + L + 5)$ is the number of iid π_0 draws scored in the final run. The exact branch (full enumeration) is taken when $\sum_{k \le K} \binom{p}{k} \le (5H_0 + 4N_0)p$.

Regimes. R1: $n \to \infty$, s_0 fixed, $p \le n^A$. R2: p fixed (SMC statements are component results; enumeration eventually applies). R3: data and $\mathcal{C}$ fixed while N or the number of islands grows (idealised). R4: the actual budgets along R1. Fixed-data statements condition on a successful frozen calibration, never on passing later diagnostics.

3 Assumptions

The full inventory, with every code's precise condition, is `theorems/00_COMMON.md` §3. We discuss the groups.

Statistical (target) assumptions. G (Gaussian fixed design) and R (fixed s_0, $p \le n^A$, bounded signal energy, eigenvalue floor on the true design) are the project's standard bundle. SE (sparse eigenvalues on every set of size $\le m_n + s_0$) is strong and cannot be checked exhaustively; it excludes exact duplicates. Beta-min comes in two forms: BM-C with an explicit fixed constant $M_* = 2L_* + 1$ obtained by auditing the project's selection proof (provisional; the proving run may replace it), and BM-∞ with $M_n \to \infty$. The sharp constant for general designs is open; in the orthogonal benchmark the transition is sharp (Conjecture 4.5).

Start-side assumptions. VIS (actual-block visibility of each true variable's block projection), $\mathrm{NL}_{\mathrm{avg}}(\kappa)$ (averaged no-leakage of signal into noise coordinates within blocks), and EXP, the uniform block-score expansion. EXP is the honest name for what the review group's "AGG-stat" silently assumed: a uniform one-sided expansion of every table addition ratio with a remainder $R_n = O_P(1)$ (or $o_P(\log n)$). Deriving EXP from G, R, SE is the main open analytic step of Part C; the judge's assistant could only observe a remainder range of $[-0.10, 0.17]$ on one small design [provisional evidence, Sol check J5].

Computational hypotheses. These are conditional obligations, never asserted properties of v3: HIT and PX (a scoring route succeeds), LK_d, EXIT_d, GAIN_d (leakage, first-exit and gain controls for the terminal repair), BASIN (abstract basin constants), COVER (full-space

coverage of the candidate set at every grid and over-tempered evaluation point), BW2 / BW_W / WS / PW2 (particle-system stability conditions of increasing specificity), CPL-E / CPL-H / RB (couplings with a reflected comparator, output-specific), MISS-E / MISS-H (hidden-set avoidance), OP-a / OP-s (operational availability), NUM, NDR, STIFF. The declared idealisations EX-A (exact arithmetic), EX-S (ideal streams) and CAL (frozen calibration) are used throughout Parts B–D.

What was retired. MASS (replaced by exact start-mass identities), ζ-EXIT (refuted on every dataset by the addition floor), AGG-stat as a hypothesis, OP's numerics clause, and the optional over-tempered ideas V$\leq$, OT-lb, NA. The regimes are not declared mutually exclusive except where it is a derivation (exact duplicates conflict with SE and NIS; ORTH forces $p \leq n-1$).

4 Part A: the target posterior

Conjecture 4.1 (A0: contrast reduction and mixed-rate transfer)**.** *[file T01] For every valid input, the v3 posterior equals the posterior of the contrast model* $y_c \mid \theta_S, v \sim N_{n-1}(X_{c,S}\theta_S, vI)$, $\theta_S \mid v \sim N(0, nv(G_S+I)^{-1})$, $v \sim \mathrm{IG}(0.51, b_0 s_y^2)$, *with* $m(S) = c_n(s_y) m^c_{b_0 s_y^2}(S)$ *for a model-independent* $c_n(s) = s^{n-1}\Gamma(0.51)/(\sqrt{2\pi b_0}\Gamma(0.01))$. *For rates* $b_- \leq b \leq b_+$, $\kappa^{-1} m^c_{b_+} \leq m^c_b \leq \kappa m^c_{b_-}$ *with* $\kappa = (b_+/b_-)^{0.51}$, *also for joint numerator integrals; under G, R the empirical rate* $b_0 s_y^2$ *lies in a compact interval with probability* $\to 1$.

Purpose. This is the foundation of every statistical transfer: the project's proofs were written for a fixed-rate target, and v3's rate is random. The transfer is legitimate only in the mixed-endpoint form (numerator at b_-, denominator at b_+). Identities verified to 10^{-14} on small data [provisional evidence, Sol check J1, J2].

Conjecture 4.2 (A1: dimension control and no strict supersets)**.** *[file T02] In R1 under G, R, for every deterministic threshold* m, $\Pi(|S| > m \mid y) \leq 2^{-m}(2p)^{s_0}(n+1)^{s_0/2}\sqrt{n}\,\Theta_{n,m}$ *with uniformly tight* $\Theta_{n,m}$; *hence* $\Pi(|S| > m_n(\omega_n) \mid y) = O_{\mathbb{P}_0}(e^{-\omega_n})$. *Under SE,* $\Pi(S \supsetneq S_0 \mid y) = O_{\mathbb{P}_0}(r_n)$.

Purpose. The operational cap $\lfloor n/2 \rfloor$ is large; this shows the posterior nevertheless lives on $O(\log n)$ models, so the cap need not be changed and no eigenvalue condition near the cap is needed.

Conjecture 4.3 (A2: selection, screening, PIPs, global HPM)**.** *[file T03] In R1 under G, R, SE and BM-C (or BM-*∞*):* $1 - \pi(S_0) = O_{\mathbb{P}_0}(r_n)$; $\max_j |\mathrm{PIP}_j - \mathbf{1}\{j \in S_0\}| \leq 1 - \pi(S_0)$; $\mathbb{P}_0(\mathrm{MPM} = S_0) \to 1$; $\mathbb{P}_0(S^\dagger \neq S_0) \leq 2\mathbb{E}_0[1-\pi(S_0)] \to 0$, *and on* $\{\pi(S_0) > 1/2\}$ *the truth is the strict global mode. Without full beta-min, the strong coordinates* J_n *are screened in:* $\Pi(S \not\supseteq J_n \mid y) = O_{\mathbb{P}_0}(r_n)$.

Purpose. The central target result, and the target half of every combined HPM theorem. The strict-mode clause is what makes the all-scored HPM decision work (Part C). The fixed constant answers "what signal strength suffices for $p > n$" conservatively; its sharp value is open.

Conjecture 4.4 (A3: contraction without beta-min)**.** *[file T04] In R1 under G, R: prediction contracts at* $\epsilon^2_{2,n} = (s_0+1)\log^2(p \vee n)/n$; *under SE the posterior dimension is bounded by* $d_* = s_0 + \lceil s_0(2A+1) \rceil$, *prediction and coefficients contract at* $\epsilon^2_{1,n} = (s_0+1)\log(p \vee n)/n$ *(coefficients with the factor* ϕ_0^{-2}*), and the contrast-scale variance satisfies* $|v/\sigma_*^2 - 1| \leq C[\epsilon_n^2 + u_n/\sqrt{n}]$ *for any* $u_n \to \infty$; *under DX the raw coefficients inherit the rate with the factor* d_-^{-2}.

Purpose. The guarantee that survives when exact selection is inappropriate (weak signals). Promoted to core by the judge because it answers a central question of the brief.

Conjecture 4.5 (A4: orthogonal threshold and benchmark gap)**.** *[file T05] Under ORTH, $D_S = n^{|S|}$ and $Q_S = y^\top y - n^{-1}\sum_{j\in S} z_j^2$; with $\lambda_n = (n-1)b_n^2/\sigma_*^2$ and $\tau_n = \log n + 2\log((2p_n - s_0)/s_0)$, if $\sqrt{\lambda_n} - \sqrt{\tau_n} \to c$ then $\mathbb{P}_0(S^\dagger = S_0) \to \Phi(c)^{s_0}$. Any signal sequence between $\sqrt{2\log(p_n - s_0)}$ and $\sqrt{\tau_n}$ (both differences diverging) is recovered by the known-size, known-noise selector while v3's global HPM fails; the gap is $\log n + O(1)$ in squared signal, the price of $g = n$.*

Purpose. Separates target-induced omission from computational failure, and quantifies a consequence of the fixed slab. Secondary because it is a benchmark ($p \le n - 1$).

Conjecture 4.6 (A5: identifiability and exact ties)**.** *[file T06] Two feasible score functions coincide for every standardised response iff $X_S X_S^\top = X_{S'} X_{S'}^\top$ (equal sizes follow from traces); under G and NIS accidental exact ties are null; without NIS, A2's strict mode makes the tie rule irrelevant at S_0. Signed duplicates make the support label unlearnable by any method.*

Conjecture 4.7 (A6: size-only ranking ceiling and Bayes decisions)**.** *[file T07] No size-only prior can repair a within-size likelihood ranking error: $\{S^\dagger = S_0\} \subseteq \{S^*_{s_0} = S_0\}$. The global HPM and the MPM are Bayes for 0–1 and Hamming loss on the target; the reported $\widehat{S}$ and estimated PIPs inherit this only with their computational guarantees.*

Conjecture 4.8 (A7: oracle joint Bernstein–von Mises limit; optional)**.** *[file T08] Under G, R, SE, BM, GL, conditional on $S = S_0$ the contrast coefficients and variance have the oracle Gaussian limit $N(0, \mathrm{diag}(\sigma_*^2\Gamma_0^{-1}, 2\sigma_*^4))$ after $\sqrt{n}$ scaling; the scaled coefficient covariance is exactly $n^2 v H_{S_0}^{-1}$.*

5 Part B: Monte Carlo validity and rates (fixed data)

Given EX-A, EX-S and CAL, *validity* is exact and needs no other condition: the evidence and the evidence-weighted Rao–Blackwell rewards are unbiased, the Markov bound holds, and every feasible Gibbs addition has an inverse-polynomial hazard floor. *Accuracy* is always conditional and passes through one of four routes: concentration of the bridge on a single model (first moments), a reflected comparator with a coupling, particle-system stability (BW2 or a working set), or a crossing benchmark. The output never verifies any of them.

Conjecture 5.1 (T7: exact start, invariant kernels, whole-round minorisation)**.** *[file T09] The convolution and backward sampler give $\bar{Z}_0$ and exact iid π_0 draws; every elementary kernel is q_ϕ-reversible including the cap boundaries; the order-averaged round is invariant and self-adjoint (a realised order need not be reversible); every realised order satisfies $K_\sigma(x,\cdot) \ge \epsilon_\phi q_\phi$ with $\epsilon_\phi = \inf_S \pi_0(S)/q_\phi(S)$, giving a TV bound $(1-\epsilon_\phi)^h$. A tiny available ϵ_ϕ is not evidence of slow mixing.*

Conjecture 5.2 (T11: feasible-addition hazards)**.** *[file T10] For $|S| = k < K$ and $j \notin S$, at every $\phi \in [0,1]$: $\gamma_\phi(S+j)/\gamma_\phi(S) \ge r_k = (k+1)/((2p-k-1)\sqrt{n+1})$, so $o_j(S,\phi) \ge h_k = r_k/(1+r_k)$ and $\pi_0(S+j) \ge r_k \pi_0(S)$. A round from S visits $S \cup J$ ($|J| = a$) with probability at least $\binom{m}{a}^{-1}\prod h_{k+i}$ (an ordering factor is unavoidable, even for $a = 1$); at an actual execution the entry probability is $o_j(S,\phi) \ge h_k$ conditional on the whole past.*

Purpose. The quantitative accessibility fact that reorganises the negative results: no mode reachable by additions is exponentially hidden; exponential protection needs deletions, swaps or the cap. The full-round ordering effect was confirmed on a complete hybrid round [provisional evidence, Sol check J8].

Conjecture 5.3 (L-LED/T14-D: the scoring and discovery ledger)**.** *[file T11] Each mechanism scores a fixed model S^* with an exact conditional probability at its execution time: Gibbs at j iff $X_{\text{pre-}j} \setminus \{j\} = S^* \setminus \{j\}$; uniform swap $1/[k(p-k)]$; neighbour swap $1/[k|N(j) \setminus S|]$; iid*

proposals $\pi_0(S^*)$ regardless of acceptance; tables and scans deterministically given the block or centre. The final iid draws miss S^* with probability $(1-\pi_0(S^*))^{Q_f}$; stopped opportunities obey $\mathbb{P}(\text{no hit}, O \geq D) \leq (1-b)^D$ and $\mathbb{P}(\text{hit}) \geq b\mathbb{E}O$; set exposure is bounded by the first-hit hazard sum. If $S^\dagger$ is scored then $\widehat{S} = S^\dagger$.

Purpose. One ledger serves the positive discovery theorems (Part C) and the negative avoidance results (Part D). HPM recovery is a scoring question, not a mixing question.

Conjecture 5.4 (T8: frozen-run identities)**.** *[file T12] Given $\mathcal{C}$: $\mathbb{E}[\widehat{Z}_{r,t}\widehat{q}_{r,t}(f)] = Z_t q_t(f)$ through terminal resampling, invariant rounds and pre-draw recording positions; $\mathbb{E}\widehat{Z}_r = Z$; $\mathbb{E}[\widehat{Z}_r \widehat{\mathrm{PIP}}_{rj}] = Z\,\mathrm{PIP}_j$; $\mathbb{E}\log\widehat{Z} \leq \log Z$; $\mathbb{P}(Z \geq \alpha\widehat{Z}) \geq 1-\alpha$. Normalised PIPs and $\log\widehat{Z}$ are not unbiased in general; nothing survives conditioning on favourable diagnostics; NUM gives no such guarantee for implemented evidence.*

Conjecture 5.5 (T10: restricted ESS and omitted mass)**.** *[file T13] With $a(\psi) = q_\psi(V)$ and $w = 2v - u$,*

$$1 + \chi^2(q_v \| q_u) = \rho_V(u,v)^{-1}\,\frac{a(v)^2}{a(u)\,a(w)};$$

with COVER at both u and the over-tempered point w, an accepted interval has $1 + \chi^2 \leq 10/(9(1-\epsilon_C)^2)$. If calibration never scored a set $\mathcal{M} \not\ni \varnothing$, the grid equals the grid of the problem restricted to $\mathcal{M}^c$.

Purpose. Says exactly what the candidate grid controls (a restricted one-step moment) and why hidden models remain invisible to it.

Conjecture 5.6 (L-FM: first-moment evidence consistency)**.** *[file T14] With $\eta_t = 1 - q_t(S^\circ)$ for a model S° fixed given $\mathcal{C}$, for every particle count,*

$$\mathbb{E}|\widehat{Z}_r/Z - 1| \leq \frac{2}{1 - \max_t \eta_t} \sum_{t=0}^{T} \eta_t,$$

also for the pooled evidence; hence summable deficits give relative consistency with no over-tempered condition and no variance bound. For a set $\mathfrak{A}$ of several models an additional within-set fluctuation term $\Phi_t \geq 0$ appears that mass concentration does not control.

Purpose. The repair of the review group's objection O3: no-switch evidence consistency at the actual budgets without the over-tempered hypothesis. Constant and endpoint term verified exactly on finite laws, where tiny deficits coexist with $\mathbb{E}(\widehat{Z}/Z)^2 \approx 10^6$ [provisional evidence, Sol check J6].

Conjecture 5.7 (L-OCC: occupancy from an evidence lower tail)**.** *[file T15] $\mathbb{E}f_r(\mathfrak{A}^c) \leq \pi(\mathfrak{A}^c)/c + \mathbb{P}(\widehat{Z}_r/Z < c)$ for the fraction of island r outside a set $\mathfrak{A}$ at stage T or at a fixed terminal prefix; the lower tail must be islandwise.*

Purpose. Breaks the v1 circularity between the evidence theorem and the PIP theorem: decision outputs need only a lower tail of the evidence.

Conjecture 5.8 (T9 and R-T13: idealised limits)**.** *[file T16, T36] With $\mathcal{C}$ frozen and $N \to \infty$, the relative evidence and PIP errors are $O_{\mathcal{C}}(N^{-1})$ in mean square with a Gaussian limit; with independent islands, the secondary pooling converges to PIP_j while the primary pooling converges to $\mathbb{E}[\widehat{\mathrm{PIP}}_{1j} \mid \mathcal{C}] \neq \mathrm{PIP}_j$ in general. With a frozen stage ceiling and $H \to \infty$, the pilot schedule equals that of the deterministic rule applied to the population ESS (under NDR).*

Conjecture 5.9 (L-RC: reflection with output-specific coupling)**.** *[file T17] Reflecting every elementary transition at a set W gives a comparator with evidence mean $Z\,q_1(W)/q_0(W)$ (not $Zq_1(W)$); a common-randomness coupling fails only through the first exit, with probability at most $4Nq_0(W^c) + \sum_e \mathbb{E}[\mathbf{1}\{\tau > e^-\}K_e(X_{e^-}, W^c)]$. Evidence needs coupling through the last weight evaluation; HPM needs every scored proposal and scan; PIPs need the recorded rewards, whose reflected mean μ_j^W differs from $\pi(j \in S \mid W)$ but satisfies $|\mu_j^W - \mathrm{PIP}_j| \le \pi(W^c)$. A large union bound proves no impossibility.*

Conjecture 5.10 (B-RP: reflected single-flip pair benchmark; secondary)**.** *[file T18] Reflected at a single-flip pair, the switching Gibbs coordinate refreshes the pair exactly each round, so the comparator's relative variance is $\prod_t(1 + \chi_t^2/N) - 1$ (divided by four over islands); with pair coverage, an explicit Chebyshev bound follows. This is a statement about the reflected process; a proxy-swap pair has no refresh.*

Conjecture 5.11 (T12: conservative working-set bounds; optional)**.** *[file T19] Under a normalised future-weight bound a on a working set, the reflected evidence and reward estimators obey $\mathbb{E}(\widehat{Z}_r^W/Z_W^\circ - 1)^2 \le (1 + a^2/N)^T - 1$ and a corresponding reward bound; actual outputs inherit probability bounds through the couplings. New judge conjecture, not numerically validated; its applicability is restrictive.*

6 Part C: combined guarantees for the actual output

Conjecture 6.1 (T15: local search)**.** *[file T20]* `local_certified` *means exactly that the returned model was scanned as a centre and had no strictly improving feasible flip or swap (weak local maximality). On the event that S_0 is the strict global mode (probability $\to 1$ by A2), every completed scan from a flip-or-swap neighbour of S_0 scores S_0; moving and certifying need a remaining move and a further scan. Under ORTH every best improvement reduces the distance to S_0 (secondary); under COH the same is an optional conjecture. Under SE alone the claim is false: a signal-bearing proxy can win the first improvement [provisional evidence, Sol check J3].*

Conjecture 6.2 (T14-H: the reported HPM)**.** *[file T21]* $\mathbb{P}\{$*no usable output or* $\widehat{S} \ne S_0\} \le \mathbb{P}(U) + 2\mathbb{E}_0[1 - \pi(S_0)] + \mathbb{P}\{U^c, S_0 \notin \mathcal{S}\}$*, and the discovery term vanishes under any one route: (a) fresh iid draws, with the launch-qualified bound $\mathbb{E}[\mathbf{1}\{$launched$, S_0 \notin \mathcal{S}_{\mathrm{cal}}\}(1 - \pi_0(S_0))^{Q_f}]$ plus the non-launch term (one cancelled variable from a start mode of bounded-below mass gives intensity of scale $N_e(T + L + 5)/(p\sqrt{n})$, diverging for $p \lesssim \sqrt{n}$); (b) a Gibbs execution next to S_0; (c) swap executions from the proxy model with a stopped opportunity count (PX); (d) tables or a scanned neighbour centre; (e) terminal repair (T18′); (f) the exact branch.*

Purpose. The area-3 headline: production HPM recovery without posterior mixing, with the hypothesis of each route named. The budget correction (a factor n in v2) is [provisional evidence, Sol check J7].

Conjecture 6.3 (T16: start aggregation and conditional concentration)**.** *[file T22] Deterministically, for any feasible reference S° with the aggregate activities $q_n < 1$, e_n: $\pi_0(S^\circ) \ge (1 - q_n)^{s+1}/(1 + e_n)^s$, and $q_n \le u_n$ (an average of maximal table addition ratios); the truth odds relative to S° are an exact product of table and prior ratios. Statistically, under G, R, within-block SE, VIS, $NL_{\mathrm{avg}}(\kappa)$* and EXP*: $1 - \pi_0(S_0) = O_{\mathbb{P}_0}(2^B\sqrt{\log n}\,n^{-\kappa} + e_n)$. Along the bridge, $1/q_\phi(S_0) - 1 \le R_0^{1-\phi}R_1^\phi$ (Hölder), so $\inf_{[0,1]} q_\phi(S_0) \ge \min\{\pi_0(S_0), \pi(S_0)\}$.*

Purpose. The one regime in which the composite start is statistically correct; it makes the analytic obligation (EXP) explicit. Deterministic bounds pass 376 exact cases [provisional evidence, Sol check J5].

Conjecture 6.4 (T18: basin to output; file T23)**.** *Assume the abstract basin condition with constants $(\mathcal{R}, r_0, a, \eta, \rho, \delta)$ (entrance within r_0 rounds with probability at least a, one-round first exit at most η, truth prefix-departure at most ρ, expected initial outside fraction at most δ). After H terminal rounds*

$$\mathbb{E} f_H(S_0^c) \le \delta + (1-a)^{\lfloor H/r_0 \rfloor} + H(\eta + \rho),$$

and the measured-round PIPs obey the pathwise bound

$$\max_j \left| \widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\} \right| \le \bar{B}_r + u_\pi$$

in terms of the fraction $\bar{B}_r$ of particle-rounds with a bad prefix; the primary average has error at most

$$\delta + (1-a)^{\lfloor L/r_0 \rfloor} + (L+4)(\eta + \rho) + u_\pi,$$

with $\rho \le (B+3)u_\pi$ by reversibility and δ from L-OCC. The single-flip pair is a conditional instance; ONE alone establishes no basin.

Conjecture 6.5 (T18′: conservative terminal repair)**.** *[file T24] For fixed d with LK_d, $EXIT_d$, $GAIN_d$: from any state in $\mathcal{R}_d$ (any subset of the truth plus at most d extras) one $\phi = 1$ round reaches S_0 with probability at least $a_n = [c_n\{1 - (s_0+d)e^{-g_n}\} - 2^{s_0+d}\Lambda_n - \rho_*]_+$, $c_n = 1/(s_0+d+B+2)!$, and first exits from $\mathcal{R}_d$ have probability at most $(B+3)2^{s_0+d}\Lambda_n + \xi_n$. Since $L \to \infty$, any fixed positive c_n suffices: an eligible particle scores S_0 with probability $\to 1$ whatever the number of cancelled true variables. The earlier slot constant was withdrawn because internal swaps can change the greedy maximiser without exiting [provisional evidence, Sol check J4].*

Conjecture 6.6 (T19: evidence through a crossing)**.** *[file T25] Under BW2 the reflected comparator has relative mean-square error at most $D_N = \prod_t (1 + (C_t - 1)/N_e) - 1$ and, with a coupling, the actual evidence inherits a probability bound. Marginal return/occupancy conditions do* not *imply any such rate: an abstract two-state system with exact coverage and lazy within-pair movement has variance far above the refresh product [provisional evidence, Sol check J10]. The missing ingredient is a squared-evidence-weighted recursion (PW2), recorded as a research target.*

Conjecture 6.7 (T17: no-switch consistency at the actual budgets)**.** *[file T26] Under G, R, SE, BM-C, VIS, NL_{avg}, EXP, the idealisations and operational completion: $\widehat{Z}/Z \to 1$ and $\log \widehat{Z} - \log Z \to 0$ in joint probability (through L-FM with $(T+1)\Delta_n \to 0$, $T = O(\log^2 n)$); primary and secondary PIPs are uniformly consistent and $\widehat{\mathrm{MPM}} = S_0$ with probability $\to 1$ (through L-OCC and truth-prefix stability); the HPM is recovered from the pilot's initial draws. No Monte Carlo rate, bounded stage count or clean-flag statement is included.*

7 Part D: limits and negative results

Conjecture 7.1 (T20-C: cancellation and the iid budget)**.** *[file T27] Along a completion chain with a two-sided sequential table-ratio law $n^{-1/2}e^{O_P(1)}$ and bounded-below reference mass, $\pi_0(S_0) = p^{-c}n^{-c/2}e^{O_P(1)}$ for c cancelled variables, and the exponent of $Q_f\pi_0(S_0)$ is $\max(1,a) - c(a + \frac{1}{2})$ at $p \asymp n^a$: one cancellation is iid-cheap up to $p \asymp \sqrt{n}$ and never for $p \ge n$. Lower bounds alone prove no expense; other scoring routes are untouched.*

Conjecture 7.2 (T21-E: evidence loss under avoidance and coupling)**.** *[file T28] If calibration never scores a hidden set $\mathcal{M}$ and the weight-bearing populations stay coupled to the comparator reflected at $\mathcal{M}^c$ (failure h_n^E), then $\mathbb{P}(\widehat{Z}/Z > x) \le h_n^E + r_W/x$ with $r_W = \pi(W)/\pi_0(W)$, so $\widehat{Z}/Z \to 0$ when $r_W \to 0$ although $\mathbb{E}\widehat{Z} = Z$; evidence-measurable diagnostics have the*

comparator's law. Loss needs $S_0 \in \mathcal{M}$, and exponential loss needs every bounded superset of S_0 in $\mathcal{M}$ (by the addition floors). No production family satisfying the hypotheses has been constructed.

Conjecture 7.3 (T21-H: HPM and PIP loss)**.** *[file T29] HPM failure needs avoidance of* every *scoring mechanism (rejected proposals and Gibbs alternatives count); PIP comparisons need the actual recorded rewards. A bounded-state additions-first path to S_0 at $\phi = 1$ is at most polynomially rare, which is accessibility, not discovery at the budget.*

Conjecture 7.4 (T21-X: pilot discovery under the repair hypotheses; optional)**.** *[file T30] Under the hypotheses of T18′, an eligible pilot particle and executed pilot terminal rounds, $\mathbb{P}(S_0 \in \mathcal{S}_{\mathrm{cal}}) \to 1$; hence no hidden set containing the truth can satisfy MISS-E or MISS-H, and no exponential evidence-loss instance exists. Bounded occupancy alone (BOCC) does not supply the repair hypotheses; the universal contrapositive is withdrawn. One actual-partition instance with two cancelled variables scored the truth in the first bridge round [provisional evidence, Sol check J12].*

Conjecture 7.5 (T22: what the certificates establish)**.** *[file T31] $0.05\widehat{Z}$ has coverage at least 95% under the ideal final-run assumptions and no accuracy content; island agreement certifies nothing over the class of unbiased estimators (a two-point law); evidence-measurable diagnostics have no power against a coupled loss instance; simultaneous discovery certificates are $\lfloor 1/t \rfloor e^{-Q_f t}$ for a fixed threshold and "every model with $Q_f \pi_0(S) \geq \log[1/(\delta \pi_0(S))]$ is scored" with probability $1 - \delta$ (substituting $\pi_0(\widehat{S})$ into a fixed-model formula is invalid); the restricted ESS controls only its restricted moment.*

Conjecture 7.6 (T23-C, T23-N, T23-S, T23-R: resources, numerics, statuses, bisection)**.** *[file T32–T35] The declared budgets give polynomial arithmetic-operation counts and branch-specific storage contracts (dense Gram $8p^2$ bytes on the SMC branch; a streaming exact branch needs $O(np + pK + nK + K^2)$ words), not wall-time or preflight guarantees; a score-error bound η preserves the ranking of models actually scored when the gap exceeds 2η (A2 supplies a $\frac{1}{2}\log n$ margin at S_0), but does not preserve an adaptive trajectory;* schedule_limit *returns the best scored model without evidence or PIPs, so HPM routes completed earlier need no schedule assumption; the universal memory ceiling $p_{\max} = \Theta(\sqrt{M_0})$ is false because enumeration precedes Gram construction; the candidate-grid bisection is obstructed only under a mass-qualified condition $\rho(a, D(b-u)2^{-60}) < 0.9$, and the discrepancy range is at most $[(J_{\max}+1)\nu + K]\log(n+1)$ (the former bound and the 10^{17} cutoff were wrong) [provisional evidence, Sol check J9].*

8 How the results fit together

Shared assumptions. G, R, SE and a beta-min form carry all of Part A and enter Part C only through A2's strict mode (T14-H, T15, T18, T18′) and endpoint concentration (T16, T17). EX-A, EX-S and CAL carry all of Parts B–D. The start-side assumptions VIS, $\mathrm{NL}_{\mathrm{avg}}$ and EXP appear only in the no-switch regime (T16, T17). The repair assumptions LK, EXIT, GAIN appear only in T18′ and its users (T14-H route (e), T21-X). Coverage, coupling and particle-stability assumptions appear only in the crossing and loss results (T10, B-RP, T12, T19, T21).

Which result feeds which. A0 → A1 → A2 is the gate for everything statistical. T7 → T8 → {L-FM, L-OCC} is the gate for everything about the final run's outputs. T11 → L-LED is the gate for discovery and avoidance. T16 + L-FM give the evidence in the no-switch regime; L-OCC + T18 give the PIPs; L-LED + A2 give the HPM. T18′ extends discovery beyond one-move starts and, through T21-X, bounds what the negative results can contain. L-RC +

T8 give the negative comparison theorems, and T22 reads every certificate through them. The dependency graph is acyclic (`THEOREM_MAP.md`).

What the theory says about the methodology. Conditionally on the transfers: the target is the right object (A1–A3, with A4's explicit price for $g = n$); the sampler is exactly valid (T7, T8); the HPM decision is robust because scoring is cheap along several routes (L-LED, T14-H, T15); the PIPs and the evidence are consistent in the no-switch regime at the declared budgets (T16, T17) and conditionally at one-move crossings (T18, B-RP); the evidence can be badly wrong with every check passing, but only under avoidance and coupling hypotheses for which no regression instance has been built (T21-E, T22); and bounded-size trajectories with the repair hypotheses exclude such instances altogether (T18′, T21-X). The main analytic obligations are EXP (start concentration), the statistical derivation of LK/EXIT/GAIN (repair), and a particle-system stability condition for crossings (PW2).

9 Recommendations to the author (not adopted)

1. Report $\pi_0(\widehat{S})$, Q_f and a multiplicity-aware discovery threshold; do not present the selected model's fixed-atom miss expression as a global certificate.
2. Consider reporting the exact start mode (max-product dynamic programme), its distance from the reported HPM, and the final search centre's distance from that HPM.
3. Make the scoring reading SR explicit in implementation tests: both feasible Gibbs alternatives and rejected proposals enter the persistent all-scored ledger.
4. Add execution-time exposure summaries where useful; round-start occupancy cannot replace them.
5. Specify branch-specific storage, cache, clustering and arithmetic-error contracts before adopting implementation complexity or numerical guarantees.
6. Correct scan and grid-cost reporting units ($\max_k k(p-k)$; log-sum-exp calls versus scalar summands).
7. If unrestricted asymptotic schedule completion becomes a requirement, investigate adaptive bisection depth as a separate revision.
8. Consider reporting completed-table activities and leakage diagnostics to assess the start assumptions.
9. Treat joint pair-addition proposals as a separately evaluated future modification.
10. Keep availability, conditional accuracy and certification separately visible in theorem statements and output documentation.

10 Provenance

Council: 2 Fable 5.1 (high) + 2 Astra (xhigh), one independent round and three joint rounds (`council/`); review group: 4 Opus 5.5 (high), two rounds and one send-back with six objections, all accepted (`review/`); judge: one Astra session, phase-1 review with 38 findings and 12 orders, two Sol 6.1 assistants (`judge/sol1`, `judge/sol2`; scripts and evidence kept there, labelled provisional), phase-2 verdict with 78 changes (`judge/PROPOSAL_v3.md`); lead's rulings in `RULINGS.md`. External primary precedents not opened in this run are marked UNVERIFIED in the files; the project's own proofs (`Proposed_project/proofs`) are transfer resources, not adopted results.